

RoboLab scenes and exact prompts

WORLD-ACTION MODEL WORKSHOP STUDY | 26 SEPTEMBER 2026

Can a world-action model's predicted future help distinguish following the wrong goal from failing to execute the right one?

A robot can fail because it acts toward the wrong goal, or because it cannot complete the intended movement. We test whether the future a world-action model generates helps distinguish those observable cases. A plausible video or a low success rate alone cannot answer this.

Read the pictures first, then compare the prompts

Different goals: the requested relation, destination or object changes. Prediction and action should respond to that change.

Same goal, different wording: D is direct, S puts the moved object first in the relation clause, and I puts the reference first using a converse relation. The desired physical result stays the same.

Nominal means exact RoboLab text. New instructions on an old scene are explicitly marked study-added goals. We do not call them native benchmark tasks.

Proposal	Scope
Primary	5 scenes / 14 physical goals / 42 exact prompts
Held scene	Mug: 9 prompts; reset validity must be resolved
Optional	5 additional object, distance or destination prompts
Current evidence	15 Nano pilot episodes; no new experiment started

Workshop fit: primary Motion 6 (what success-only benchmarks miss), secondary Motion 4 (whether model predictions can support evaluation). This study does not establish that world models are necessary or better than policies without exposed predictions.

Reference frame: robot frame. Images show actual initial shoulder-camera inputs; the wrist input is also used by the policy. Native scene backgrounds are retained. Proposed prompt interventions are not yet qualified or run.

S1. Cube, bowl and banana

rubiks_cube_banana_bowl.usda | core



First raw policy observation, before the first action. Two original shoulder views; not a generated scene illustration.

RoboLab nominal prompts

“Put the rubiks cube to the left of the bowl”

“Put the rubiks cube in front of the bowl”

“Put the rubiks cube behind the bowl”

What we test: Does the predicted and executed cube placement follow four distinct directions, and survive an equivalent reversal of the reference wording?

Keep in mind: The stock RubiksCubeRightOfBowlTask loads rubiks_cube_banana_bowl_mug_bin.usda, a DIFFERENT scene. We use the simple scene shown here for all four goals. Right is a study-added goal on this scene; it is not a paired run of that stock task.

Four goals on this fixed scene: left, right, in front, behind. The next page gives all twelve exact instructions.

S1. Four goals, three descriptions each

Cube left of bowl (native on this scene)

ID / form	Exact instruction
S1-L-D	Put the rubiks cube to the left of the bowl
S1-L-S	Place the Rubik's cube so that the Rubik's cube is to the left of the bowl.
S1-L-I	Place the Rubik's cube so that the bowl is to the right of the Rubik's cube.

Cube right of bowl (study-added goal; native right wording reused on fixed simple scene)

ID / form	Exact instruction
S1-R-D	Put the rubiks cube to the right of the bowl
S1-R-S	Place the Rubik's cube so that the Rubik's cube is to the right of the bowl.
S1-R-I	Place the Rubik's cube so that the bowl is to the left of the Rubik's cube.

Cube in front of bowl (native on this scene)

ID / form	Exact instruction
S1-F-D	Put the rubiks cube in front of the bowl
S1-F-S	Place the Rubik's cube so that the Rubik's cube is in front of the bowl.
S1-F-I	Place the Rubik's cube so that the bowl is behind the Rubik's cube.

Cube behind bowl (native on this scene)

ID / form	Exact instruction
S1-B-D	Put the rubiks cube behind the bowl
S1-B-S	Place the Rubik's cube so that the Rubik's cube is behind the bowl.
S1-B-I	Place the Rubik's cube so that the bowl is in front of the Rubik's cube.

S2. Mustard and two bins

two_bin.usda | core

over_shoulder_left_camera

over_shoulder_right_camera



First raw policy observation, before the first action. Two original shoulder views; not a generated scene illustration.

RoboLab nominal prompts

“Put the mustard in the left bin”

“Put the mustard in the right bin”

What we test: Does language select the correct destination, and does the predicted transfer actually end in containment?

Keep in mind: Left and right identify the bins at reset in the robot frame. Both goals use the same physical reset. The pilot moved mustard in opposite directions but missed both bins.

Mustard in left bin (native)

ID / form	Exact instruction
S2-L-D	Put the mustard in the left bin
S2-L-S	Place the mustard so that the mustard is inside the left bin.
S2-L-I	Place the mustard so that the left bin contains the mustard.

Mustard in right bin (native)

ID / form	Exact instruction
S2-R-D	Put the mustard in the right bin
S2-R-S	Place the mustard so that the mustard is inside the right bin.
S2-R-I	Place the mustard so that the right bin contains the mustard.

S3. Butter box and raisin box

butter_raisin_box.usda | core



First raw policy observation, before the first action. Two original shoulder views; not a generated scene illustration.

RoboLab nominal prompt

“Pick up the butter box and place it on top of the raisin box”

What we test: Does the model distinguish supported placement on the box from placement beside it, and keep the intended object fixed under reversed wording?

Keep in mind: On top means supported placement, not hovering above. Left/right are new goals on the same existing scene; their physical feasibility and scoring must be checked before collection. This is not a pure height ablation: support and contact also change.

Butter box on raisin box (native)

ID / form	Exact instruction
S3-TOP-D	Pick up the butter box and place it on top of the raisin box
S3-TOP-S	Place the butter box so that the butter box is on top of and supported by the raisin box.
S3-TOP-I	Place the butter box so that the raisin box is underneath and supporting the butter box.

Butter box left of raisin box (study-added goal)

ID / form	Exact instruction
S3-L-D	Place the butter box on the table to the left of the raisin box.
S3-L-S	Move the butter box on the table so that the butter box is to the left of the raisin box.
S3-L-I	Move the butter box on the table so that the raisin box is to the right of the butter box.

Butter box right of raisin box (study-added goal)

ID / form	Exact instruction
S3-R-D	Place the butter box on the table to the right of the raisin box.
S3-R-S	Move the butter box on the table so that the butter box is to the right of the raisin box.
S3-R-I	Move the butter box on the table so that the raisin box is to the left of the butter box.

S4. Mustard bottle and raisin box

mustard_raisin_box.usda | core



First raw policy observation, before the first action. Two original shoulder views; not a generated scene illustration.

RoboLab nominal prompt

“Place the mustard on the raisin box.”

Native text has one trailing space after the period; preserved in the JSON.

What we test: Does the model distinguish supported placement on the box from placement beside it, and keep the intended object fixed under reversed wording?

Keep in mind: On top means supported placement, not hovering above. Left/right are new goals on the same existing scene; their physical feasibility and scoring must be checked before collection. This is not a pure height ablation: support and contact also change.

Mustard bottle on raisin box (native)

ID / form	Exact instruction
S4-TOP-D	Place the mustard on the raisin box.
S4-TOP-S	Place the mustard bottle so that the mustard bottle is on top of and supported by the raisin box.
S4-TOP-I	Place the mustard bottle so that the raisin box is underneath and supporting the mustard bottle.

Mustard bottle left of raisin box (study-added goal)

ID / form	Exact instruction
S4-L-D	Place the mustard bottle on the table to the left of the raisin box.
S4-L-S	Move the mustard bottle on the table so that the mustard bottle is to the left of the raisin box.
S4-L-I	Move the mustard bottle on the table so that the raisin box is to the right of the mustard bottle.

Mustard bottle right of raisin box (study-added goal)

ID / form	Exact instruction
S4-R-D	Place the mustard bottle on the table to the right of the raisin box.
S4-R-S	Move the mustard bottle on the table so that the mustard bottle is to the right of the raisin box.
S4-R-I	Move the mustard bottle on the table so that the raisin box is to the left of the mustard bottle.

S5. Two bowls

bowls_2_table.usda | core



First raw policy observation, before the first action. Two original shoulder views; not a generated scene illustration.

RoboLab nominal prompts

“Stack the left bowl on the right bowl”

“Stack the right bowl on the left bowl”

What we test: When the two objects look alike, does the model predict and move the requested bowl, or act on the other one?

Keep in mind: Bind left/right bowl identities at reset, using the robot frame. In the recorded layout left is bowl_2 and right is bowl_1; recompute bindings after any layout change. Opposite stacking goals change BOTH mover and support. They are not equivalent descriptions and are not a pure direction ablation.

Left bowl onto right bowl (native)

ID / form	Exact instruction
S5-LR-D	Stack the left bowl on the right bowl
S5-LR-S	Move the left bowl so that the left bowl is stacked on the right bowl.
S5-LR-I	Move the left bowl so that the right bowl is underneath and supporting the left bowl.

Right bowl onto left bowl (native)

ID / form	Exact instruction
S5-RL-D	Stack the right bowl on the left bowl
S5-RL-S	Move the right bowl so that the right bowl is stacked on the left bowl.
S5-RL-I	Move the right bowl so that the left bowl is underneath and supporting the right bowl.

S6. White mug and table center

objects_around_table.usda | hold; excluded from primary study



First raw policy observation, before the first action. Two original shoulder views; not a generated scene illustration.

RoboLab nominal prompt

“Put the white mug in the center of the table.”

What we test: Can the model target a region instead of another object, and preserve that target across descriptions?

Keep in mind: ON HOLD. In the pilot, the bowl and banana were already far below the table at the first policy observation. Keep the 1/1 native success as a pilot result with this caveat. No confirmation episodes until reset validity is repaired and versioned. Image shows the actual incomplete pilot reset.

Mug at center (native)

ID / form	Exact instruction
S6-C-D	Put the white mug in the center of the table.
S6-C-S	Place the white mug so that the white mug rests at the center of the table.
S6-C-I	Place the white mug so that the center of the table is directly underneath the white mug, with the mug resting on the table.

Mug left of center (study-added goal)

ID / form	Exact instruction
S6-L-D	Place the white mug on the table to the left of its center.
S6-L-S	Move the white mug on the table so that the white mug is to the left of the table center.
S6-L-I	Move the white mug on the table so that the table center is to the right of the white mug.

Mug right of center (study-added goal)

ID / form	Exact instruction
S6-R-D	Place the white mug on the table to the right of its center.
S6-R-S	Move the white mug on the table so that the white mug is to the right of the table center.
S6-R-I	Move the white mug on the table so that the table center is to the left of the white mug.

Optional tests and interpretation rules

S1-X-NEAR

“Move the Rubik’s cube onto the table so that it is closer to the robot base than the bowl is.”

New radial-distance goal. Not an assumed synonym for in front of; requires its own distance predicate.

S1-X-FAR

“Move the Rubik’s cube onto the table so that it is farther from the robot base than the bowl is.”

Opposite radial-distance goal. Hold the bowl fixed; use robot-base distance, not camera depth.

S1-X-OBJECT

“Put the banana to the left of the bowl.”

Change the manipulated object while preserving the left-of relation. Requires banana feasibility and object-specific scoring.

S2-X-BETWEEN

“Place the mustard on the table between the two bins.”

New destination, not a paraphrase of either bin task. Tests two-reference localization; requires a predefined bounded between region.

S6-X-OBJECT

“Put the Rubik’s cube in the center of the table.”

Change the manipulated object while preserving the destination; held with this entire scene.

Do not collapse these distinctions

- “Not left” does not uniquely mean right. We exclude negation from the primary matrix.
- “Farther from the robot base” is radial distance; “behind” is a directional relation. These are separate tests.
- “On top” requires support; “above” can permit hovering. RGB predictions cannot prove contact or stable support.
- Bowl role reversal changes which object moves. It is a different goal, not a paraphrase.
- A goal already true at reset is a maintenance case. Report it separately from achieving a new goal.
- A generated future can be ambiguous, occluded or too short to show the decision. Mark it unknown; do not count it as correct or wrong.

Sources and exact machine-readable strings

Native source: NVlabs/RoboLab commit 0aef241fb088ca21bb4ebd24448940ed56620d17. Scene images: September 26 workstation pilot, request 000 raw observations. All 29 spatial task definitions are preserved in native_spatial_inventory.json.

Editable source: docs/roboLab-workshop-20260926/SCENES_AND_PROMPTS.md

Exact prompts and hashes: docs/roboLab-workshop-20260926/prompt_matrix.json

Workshop: <https://do-robots-need-world-models.github.io/>